

CENTRAL IT PROCUREMENT CELL - NATIONAL INFORMATICS DIVISION

NOTICE INVITING TENDER: NID/IT-NET/2026/09

TENDER FOR: Procurement of 24-Port Managed Gigabit Ethernet L2/L3 Network Switches

SECTION 2: TECHNICAL SPECIFICATIONS FOR MANAGED NETWORK SWITCHES

1. PORT DENSITY & INTERFACES:

- 1.1 Minimum 24 auto-sensing 10/100/1000 Base-T RJ-45 Gigabit Ethernet ports.
- 1.2 Minimum 4 dedicated 1G/10G SFP+ optical uplink transceiver slots.
- 1.3 Dedicated out-of-band Gigabit Ethernet management port and RJ-45/USB Type-C console interface.

2. PERFORMANCE & SWITCHING FABRIC:

- 2.1 Switching capacity shall be minimum 128 Gbps non-blocking wire-speed fabric.
- 2.2 Packet forwarding rate shall be not less than 95 Mpps (Million packets per second).
- 2.3 Packet buffer memory shall be minimum 4 MB dynamically shared across all ports.
- 2.4 MAC address table capacity: Minimum 16,000 MAC address entries.

3. LAYER-2 & LAYER-3 FEATURES:

- 3.1 IEEE 802.1Q VLAN tagging with support for minimum 4094 active VLAN IDs.
- 3.2 Support for Port-based VLAN, MAC-based VLAN, Protocol-based VLAN, and Voice VLAN.
- 3.3 Spanning Tree Protocols: Conformance to IEEE 802.1D STP, IEEE 802.1w Rapid Spanning Tree (RSTP), and IEEE 802.1s Multiple Spanning Tree (MSTP).
- 3.4 Quality of Service (QoS): IEEE 802.1p CoS with minimum 8 hardware priority queues per port, Weighted Round Robin (WRR) and Strict Priority (SP) scheduling.
- 3.5 Link Aggregation: IEEE 802.3ad LACP supporting up to 8 link aggregation groups.

4. SECURITY & ACCESS CONTROL:

- 4.1 IEEE 802.1X Port-Based Network Access Control with RADIUS and TACACS+ centralized authentication.
- 4.2 Port Security, DHCP Snooping, Dynamic ARP Inspection (DAI), and IP Source Guard.
- 4.3 Hardware Access Control Lists (ACLs) supporting IPv4 and IPv6 packet filtering based on Layer 2/3/4 headers.

5. NETWORK MANAGEMENT & MONITORING:

- 5.1 Secure Web GUI (HTTPS), Command Line Interface (CLI) over SSHv2, and Telnet.
- 5.2 SNMPv1, SNMPv2c, and SNMPv3 support with RMON groups 1, 2, 3, and 9.
- 5.3 Full dual-stack IPv4 and IPv6 management and routing capabilities.

6. POWER SUPPLY & MECHANICAL:

- 6.1 Dual redundant hot-swappable internal power supplies operating on 100-240V AC, 50/60 Hz.
- 6.2 Standard 19-inch 1U rack-mountable sheet steel chassis including complete mounting brackets.
- 6.3 Enclosure protection rating shall be minimum IP20.
- 6.4 Operating temperature: 0°C to 50°C, Operating humidity: 10% to 90% non-condensing.

7. COMPLIANCE & STANDARDS:

- 7.1 Mandatory BIS Compulsory Registration Scheme (CRS) certificate under MeitY Electronics and IT Goods Order per IS 13252 (Part 1) or IS/IEC 13252-1.
- 7.2 Electromagnetic Compatibility (EMC): Conformance to IS/IEC 61000-4-2 (Electrostatic Discharge), IS/IEC 61000-4-3 (Radiated RF), and IS/IEC 61000-4-4 (Conducted RF).
- 7.3 3 Years Comprehensive 24x7 OEM on-site warranty with advance hardware replacement SLA.